LAB ASSIGNMENT – 2.5

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BATCH - 29

Task 1: Refactoring Odd/Even Logic (List Version)

❖ Scenario:

You are improving legacy code.

❖ Task:

Write a program to calculate the sum of odd and even numbers in a list, then refactor it using AI.

❖ Expected Output:

❖ Original and improved code

Task 2: Area Calculation Explanation

❖ Scenario:

You are onboarding a junior developer.

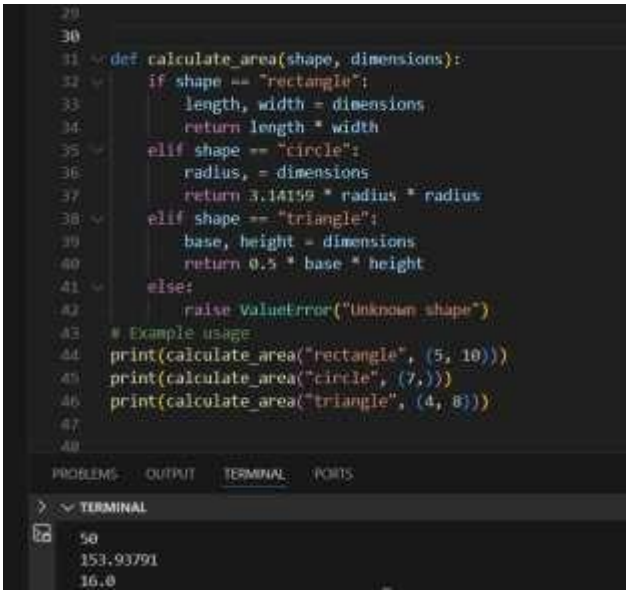
❖ Task:

Ask Gemini to explain a function that calculates the area of different shapes.

❖ Expected Output:

➢ Code

➢ Explanation



Task 3: Prompt Sensitivity Experiment

❖ Scenario:

You are testing how AI responds to different prompts.

❖ Task:

Use Cursor AI with different prompts for the same problem and observe code changes.

❖ Expected Output:

➢ Prompt list

➢ Code variations

```
def calculate_factorial(number):
    factorial_result = 1
    for i in range(1, number + 1):
        factorial_result *= i
    return factorial_result
# Example usage
print(calculate_factorial(5))
```

PROBLEMS OUTPUT TERMINAL PORTS

TERMINAL

120

Task 4: Tool Comparison Reflection

❖ Scenario:

You must recommend an AI coding tool.

❖ Task:

Based on your work in this topic, compare Gemini, Copilot, and Cursor AI for usability and code quality.

❖ Expected Output:

Short written reflection